

L09 — Exploring Image Generation Platforms in Computer Vision

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Note on this submission

The images generated during testing were not saved, so this report covers the platform research and comparative technical/ethical analysis in full, plus general qualitative impressions from testing, but does **not** include the generated images with captions that the assignment also asks for. That's a real gap, not something worked around — flagged here rather than hidden.

Introduction

Three platforms were explored: **Runway ML**, **Google Gemini** (image generation), and **NightCafe Studio**. NightCafe replaced Lobe.ai, which was part of the original research plan but is a discontinued product (Microsoft deprecated it in January 2023) and, more fundamentally, isn't a generative image tool at all — it's a no-code classifier-training tool, so it couldn't have produced comparable generated-image output anyway. NightCafe is a live, actively maintained consumer image-generation platform, making it a better fit for a real head-to-head comparison.

Platform Research

Runway ML

Runway AI, Inc. (founded 2018) builds on latent diffusion — its researchers co-developed the original Stable Diffusion architecture with LMU Munich's CompVis group in 2022. Its model line evolved through Gen-1/Gen-2, Gen-3 Alpha (June 2024), and Gen-4 (March/April 2025), which focused on "World Consistency" — generating consistent characters, objects, and environments across scenes. Gen-4.5 (December 2025) runs on an Autoregressive-to-Diffusion (A2D) architecture, blending diffusion's visual fidelity with autoregressive language/scene understanding, using Temporal Attention Layers so each frame stays contextually aware of the previous one. Runway has not disclosed full training data or parameter counts.

Primarily a video-generation platform (Runway is valued after raising \$860M from Google, NVIDIA, and General Atlantic, with partnerships including Lionsgate, AMC Networks, and IMAX), though it also offers image generation and editing. Strength: strong character consistency and micro-expression capture. Limitation: diffusion-based generation means identical prompts don't

guarantee identical outputs, and base resolution (720p, with a separate 4K upscale) lagged some 2025-era competitors.

Documented bias/ethical concern: A leaked internal spreadsheet showed Runway had scraped over 3,900 YouTube channels for training data without permission, prompting a class-action lawsuit (with Stability AI and DeviantArt) that a judge allowed to partially proceed in August 2024, plus a separate 2026 YouTuber lawsuit alleging violation of YouTube's terms of service.

Google Gemini (Image Generation)

Current Gemini image models ("Nano Banana" / Gemini 2.5+ Flash Image / Gemini 3 Pro Image) use a Multimodal Diffusion Transformer (MDT): text, uploaded images, and voice are all tokenized into one unified sequence, replacing the older pipeline where a separate language model handed a static embedding to a separate diffusion model. Gemini 2.5 Flash Image scales from 450M to 8B parameters depending on task, using a sparse mixture-of-experts approach. Google's earlier Imagen line (2022) used a Transformer architecture emphasizing photorealism; current Imagen 4 remains text-to-image-only, distinct from the multi-turn, editable Nano Banana models. Path: Imagen (2022) → Imagen 2 (Dec 2023) → Gemini native image generation (2025) → Gemini 2.5 Flash Image → Gemini 3 Pro Image (Nov 2025).

Documented failure modes — subject drift across turns, counting errors, and compositional conflict (trouble satisfying multiple constraints in one prompt) — are described by reviewers as architectural rather than incidental.

Documented bias/ethical concern (the most heavily documented of the three): In February 2024, Gemini's image generation produced racially diverse depictions of historically specific groups (the Founding Fathers, WWII-era German soldiers) when asked for literal historical imagery, along with other flagged outputs like Black Vikings and a female Catholic pope. Google's SVP acknowledged the tuning meant to ensure a range of people failed to account for cases that should clearly not show a range. The incident triggered a reported \$90B market-value loss and a temporary suspension of people-generation. A secondary pattern reported: the model would generate images of some racial groups but refuse equivalent requests for others, citing stereotype-reinforcement concerns. As of February 2026, a "Nano Banana 2" safety update further tightened restrictions on public figures, face-swapping, and photorealistic real people.

NightCafe Studio

NightCafe is a consumer-facing AI art generation platform supporting multiple underlying model backends (including Stable Diffusion-family models), with a community/social layer (public feeds, challenges, voting) distinguishing it from single-model tools like Runway or Gemini. It's positioned as an accessible entry point for AI art generation rather than an enterprise or research-lab product

— free-credit-based access, no coding or account-heavy setup required, which made it the most immediately usable of the three platforms tested.

Cross-Platform Technical Comparison

Aspect	Runway ML	Gemini (image)	NightCafe Studio
Task	Text/image-to-video (+ image gen/edit)	Text/image-to-image generation & editing	Text-to-image generation (multi-backend)
Core architecture	Latent diffusion → Autoregressive-to-Diffusion (A2D)	Multimodal Diffusion Transformer, mixture-of-experts	Diffusion (multiple selectable backend models)
Training data disclosed?	No (disputed — YouTube scraping allegations)	Partially (web docs, code, image/audio/video; June 2025 cutoff)	Varies by selected backend model
Development status	Actively developed, frontier lab	Actively developed, frontier lab	Actively developed, consumer platform
Primary output	Video (and some images)	Images	Images

General Observations from Testing

Testing confirmed the core premise behind picking three architecturally different platforms: each one expressed the same prompt in a genuinely different way rather than converging on a similar "default" interpretation — consistent with each platform running different underlying architectures and training data, exactly as the technical research above would predict. Detail level was also not uniform across platforms — some outputs were noticeably more detailed/refined than others on the same prompt, though the specific per-prompt breakdown wasn't recorded in enough detail to score formally (see note at top).

Reflection on the Bias-Probe Prompts

The tiered prompt set deliberately included two bias-probe prompts: a gendered professional-pairing prompt ("a female surgeon and a male nurse...") and a prompt structurally similar to the one that triggered Gemini's real February 2024 controversy ("Founding Fathers of AI... as if for an official 19th-century painting"). These were included specifically to test whether current safeguards (post-2024, following Gemini's public failure) over-correct, under-correct, or hold steady on prompts of this type. This is the most pedagogically important part of the assignment's design,

and it's also the part where testing notes weren't specific enough to report a confident finding — a genuine gap in this submission, not a result being omitted.

Personal Insights

The three-platform spread (a frontier video-first lab, a frontier multimodal search-company lab, and a consumer community-driven tool) turned out to be a good design choice precisely because it wasn't three similar products — Runway, Gemini, and NightCafe sit in genuinely different product categories, which is exactly what surfaced the "different platform, different interpretation" pattern observed during testing.

Conclusion

Text-to-image generation in 2026 isn't a single technology — it's several different architectural approaches (latent diffusion, autoregressive-to- diffusion hybrids, multimodal diffusion transformers) converging on similar-looking consumer products while making different tradeoffs underneath: video-first vs. image-first design, disclosed vs. undisclosed training data, and — as Gemini's 2024 incident shows — real, publicly documented consequences when bias-mitigation tuning gets it wrong in either direction.

References

- Runway AI, Inc. product/architecture documentation and press coverage (Gen-3 Alpha, Gen-4, Gen-4.5 releases).
- Google DeepMind — Gemini/Imagen model documentation; coverage of the February 2024 image-generation controversy.
- NightCafe Studio platform documentation.

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